

Teaching Statement

In this document, I give an overview of my teaching and curricular design experience and state my perspectives on course design and on the integration of artificial intelligence in teaching. I furthermore propose how I believe my teaching could fit with the curricula at EPFL and attach an overview of my most relevant courses taught.

Teaching and Curricular Design Experience

I have been enjoying teaching ever since beginning my undergraduate studies, first as a student assistant at ETH Zurich (2008-2010), and later as head assistant during my PhD studies (2010-2014). I strengthened my theoretical basics of effective teaching, tutoring, and coaching on different educational levels through a postgraduate degree in didactics from ETH Zurich (*Master of Advanced Studies in Secondary and Higher Education / Swiss Teaching Diploma*). These studies also involved two internships at secondary schools in Switzerland and allowed me to participate in outreach events by ETH Zurich at Swiss schools (e.g., *ETH Unterwegs*) where we informed and enthused students about STEM opportunities at ETH, specifically focusing on increasing diversity on the student body of Computer Science at ETH.

Since finishing my PhD, I have been the primary instructor for several courses at the *University of California at Berkeley* (2015-2016), *ETH Zurich* (2016-2017), *Graz University of Technology* (2017), the *University of Zurich* (2021-2022), the *University of Bologna* (2023), the *University of St.Gallen* (since 2018), and an invited lecturer at *Massachusetts Institute of Technology* (2017). I have furthermore been active in teaching Continuing Education courses at the *University of St.Gallen* (since 2019) and at *EPFL* (2025), and taught at several Summer Schools. At UC Berkeley and based on the student evaluations for my course there, I was offered a position as *Lecturer for Information Management* in 2016, however had to decline this position due to visa difficulties. At the *University of Bologna* and as part of a sabbatical in spring 2023, I advised on the implementation of the university's new Master program in *Digital Transformation Management* while teaching the first iteration of a mandatory first-semester course in that program. At the *University of St.Gallen*, as one of the four founding professors of the School of Computer Science, as *Academic Director* of our B.Sc. and M.Sc. programs (2022-2024), and as current *Dean*, I significantly contributed to the conceptualization, design, implementation, and delivery of Ph.D. (2019), M.Sc. (2021; reform 2024), and B.Sc. (2022) programs in *Entrepreneurial Computer Science*, which integrates technical Computer Science with skills and acumen for business. These programs were designed purposefully to leverage the University of St.Gallen's setup as a business school for a societally-oriented technical Computer Science. In these roles, I was and still am directly responsible for key decisions on program structure, regulatory setup, and institutional, political, cultural, and legal embedding. I have also been central to the definition of, application for, and recruitment of ten further faculty positions that are tailored to our designed programs and fit the University of St.Gallen's context.

Beyond Computer Science, I am currently a member of the *Extended Contextual Studies Board* which coordinates all interdisciplinary courses at the University of St.Gallen, and have recently been invited to join a university-wide board of ten "particularly engaged lecturers" to define the competencies that should shape our university's graduates over the coming decade, specifically with respect to new AI methods.

Course Design

Already since early in my teaching career, I have not been treating courses as stand-alone. Rather, I am always curious in what my students learn in parallel to attending my own courses.

Course Coordination This stance has especially been shaped by my experience conceptualizing and implementing the Computer Science curricula at the University of St.Gallen. I consciously take care that my courses fit with the department's curricular strategy, and have actively driven and contributed to the establishment of program-wide learning goals and their mapping to individual courses. I furthermore actively manage interfaces of my own courses with others in the curriculum, pointing students at related concepts in other courses and

inviting them to bring in what they have learned elsewhere. This is effortful, sometimes requiring week-by-week coordination with other lecturers, but I find that the value for students outweighs this effort.

In-Person Teaching I am an outspoken supporter of in-person teaching whenever physical co-presence has a chance to contribute to the value of a course (for the students and the lecturer). Correspondingly, and since I reject working with coercive presence pressure to attend courses in higher education, I design my courses in a way that students perceive enough value such that they *want to* attend them. This means that I design the lectures themselves in a way that does not replace the students' learning from textbooks and articles, and I communicate this to the students explicitly. My goal when meeting the students is to motivate, to structure, to show demonstrations, to point out connections among sub-topics, to contextualize with the assignments, to connect with recent developments, to respond to questions, and to create discussion. This works better when students are already prepared for a lecture, and they soon notice this, leading to positive feedback effects that sometimes make the lecture a truly great experience for everyone—such as in a recent instance in the current semester where, with my first-semester B.Sc. students in Computer Systems, we departed from the core material (on Shannon's connection of Propositional Calculus with Symbolic Relay Analysis) to discuss the nature of computation and automation, linking with topics the students were seeing in the same week in their Theoretical Computer Science course and (of course) with generative AI topics and the perception of generative AI in society; about a quarter of the students stayed over their full lunch break to keep discussing, despite my flagging that none of the topics were “exam-relevant”.

Theory and Practice In my teaching, I focus not only on theoretical knowledge that I compile from textbooks, research articles, first-hand experience, and popular media, but also give students the opportunity to gain hands-on experience in the subject matter. Practical exercises and assignments that are dedicated to specific aspects of the material at hand are suitable for this purpose, but ideally it takes the form of a more comprehensive, multi-week project that students work on semi-independently and that covers and integrates multiple aspects of the coursework. I find that this approach helps students beyond the additional practical know-how: to solidify theoretical concepts by creating something tangible—software and/or hardware—that they can relate to. I also consciously use the flexibility that the legal context of a course affords me to tune incentives (grading, etc.) to the goals I want to achieve: If I want students to deepen their understanding of a method (e.g., of linked data ecosystems as part of a *Distributed Systems* course) while working on an assignment, I will generously reward the effort they put into this assignment by coupling it with a grading element; conversely, if I want students to merely “try out” a new technology (e.g., gaze tracking as part of a *Ubiquitous Computing* course), I will use binary grading for this assignment to make sure that they rapidly move on to the next assignment after accomplishing the core task.

Reflection and Context I like to challenge students into adopting a healthily critical societal viewpoint on the technical subjects at hand. At UC Berkeley, I have successfully used (ungraded but mandatory) “opinion papers” for this purpose. Students submitted these short papers about research articles that discuss the main subject of a given week's class. Writing the papers did not only help to prepare students for class, but it also contributed perspectives for me to pick up during the lecture and to stimulate valuable discussions with and among the students – in particular about broader topics such as socio-cultural change owing to Ubiquitous Computing and critical topics such as security and privacy concerns in Internet of Things systems. Recently, in my Advanced Software and Systems Engineering course at the University of St.Gallen, I have extended an assignment on interoperability (in the sense of an IETF standardization process) by introducing a role-play element where the students assume the perspective of different stakeholders in Internet protocol standardization—from hyper-scalers to social media platforms to privacy advocacy groups—such that they experience how even seemingly pure technical processes are sometimes highly politicized.

Courses and Research Focusing on classes in the M.Sc. curriculum, I am proud that I have been successful in interlinking research and teaching not only in the context of theses, but also within curricular courses. Specifically, my M.Sc. course Ubiquitous Computing has, over the past two years, led to the publication of eight course and student projects (7x first authorship by a course-enrolled student), as short/demo papers (ACM CHI Late-breaking Work [1, 3, 4, 5]), poster contributions (ACM UbiComp Posters [6, 7, 8]), and a full paper (ACM ETRA [2]). Where possible (budget) and sensible (travel distance), I also offered the students the possibility to visit the conference and present their works themselves, which two of them have described as the “defining experience” of their Master studies.

Teaching Portfolio and EPFL

While the teaching of focused advanced courses on the M.Sc. level is clearly more conducive to advancing research projects in my group and to discovering potential candidates for Master Theses, I do enjoy teaching introductory courses a lot. I see these as a chance to sustainably shape the perspective of young students on Computer Science. I adapt my teaching methods, where my portfolio today stretches courses from B.Sc. to Ph.D., from BA to MA, and from seminars with three students to lectures with 700. Although I am somewhat skeptical about the quality of student evaluations as a proxy of course quality, I am proud that my courses receive very positive student feedback and evaluations and that, in 2024, I was nominated as one of three finalists (among more than 800 candidates) for the *Best Teaching Award* at the University of St.Gallen, despite my courses being perceived as content-dense, effortful, and “hard”.

At EPFL, I would enjoy still teaching a variety of courses, from small to large; content-wise, I would like to support EPFL’s Bachelor programs in *Computer Science*, *Communication Systems*, or *Electrical and Electronic Engineering*, possibly with introductory courses on computer systems, computer networks, distributed systems, and human-computer interaction.¹ With respect to EPFL’s Master programs, I would very much like to contribute to the *Master of Computer Science* (where I see that the specializations are currently in flux), the *Master in Communication Systems*, and the *Master in Robotics*, on topics in the fields of advanced (distributed) systems engineering, autonomous agents and multi-agent systems, human-AI interaction and, of course, Ubiquitous Computing. If desired, I would furthermore like to bring my interdisciplinary experience from research into teaching, possibly in (joint) courses with Robotics, Life Science Engineering, Environmental Science and Engineering, the Humanities and Social Sciences, and the Management, Technology and Entrepreneurship programs.

Artificial Intelligence in Teaching

Over the past years, I have embraced (generative) AI in teaching in several different forms, and I continue experimenting with how AI can enrich university teaching in-line with my teaching approach. In all my current classes, I actively discuss the utility of AI-assisted programming and contrast its different forms, from “Vibe Coding” to “Vibe Engineering”² while giving my opinion about which variant I view as most conducive to the course objectives and encouraging the students to experiment with current tools and methods (e.g., parallel coding with several Codex CLI agents). When I perceive this to bring value to a course, I furthermore sometimes make (a bit more light-hearted) tactical use of AI tools to create podcasts that summarize course aspects. All these elements contribute to open discussions on the use of this approach and thereby help me avoid situations where students feel that they need to hide their usage of AI tools—on this level, I treat this very similar to teamwork: I encourage my students to work together as long as everyone’s contributions are clearly acknowledged, and there are no penalties for such teamwork; to assess individual learning progress, I make use of exams—oral, whenever possible.

¹I speak French at an intermediate level but without much practice in the past years. If required, I estimate that I would be ready to teach courses in French after one year of living in Lausanne.

²I use terminology proposed by Simon Willison, see <https://simonwillison.net/2025/Oct/7/vibe-engineering/>

Relating to curricular design, I am a member of the group that currently plans the adaptation of our B.Sc. curriculum to more systematically cover AI-assisted programming and engineering over the first two years of the program. Beyond higher education, I have counseled the Department of Education of the Canton of St.Gallen on the integration of AI in secondary school curricula, where my main suggestion is to avoid creating a separate subject on “Artificial Intelligence” (which is very attractive for political reasons). Rather, I counsel that AI should be integrated across current secondary school subjects, from obvious integrations with mathematics (which has the potential to make mathematics more attractive to teenagers) and languages (on the development and functions of language, and next-token predictability from this perspective) to less-obvious discussions in history (automation and labor, Luddites, etc.).

Courses and Evaluations

Below, I have compiled a list of my most relevant courses taught since I assumed my current position. I have not included courses that I only taught for one semester or courses where I do not have formal evaluation data. For each course, I report the most recent overall course evaluation (1 = Very Good; 5 = Very Poor) of the course and the most recent student-reported difficulty level of the course (1 = Very High; 5 = Very Low) for context.

Name	Program	Years	# Students	Overall Evaluation	Perceived Difficulty
Computer Systems	B.Sc. CS	since 2022	40	1.20	2.30
Computer Networks (my share: 50%)	B.Sc. CS	since 2022	40	1.40	2.30
Ubiquitous Computing	M.Sc. CS	since 2022	7	1.60	2.00
Advanced Software and Systems Engineering (my share: 50%)	M.Sc. CS	since 2021	40	2.50	1.70
Computer Networks and Distributed Systems	BA Business Administration	2018–2021	25	1.67	1.83
Capstone Project En- trepreneurial Informatics	BA Business Administration	2020–2022	30	1.50	1.33
Fundamentals of Com- puter Science (my share: 12.5%)	BA Business Administration	since 2019	500 per semester (2× per year)	3.00	1.80
Fundamentals of Com- puter Science (my share: 50%)	BA Interna- tional Affairs	2019–2022	40	2.80	1.80
Introduction to Computer Systems and Networks	MA Business Innovation	2018–2022	20	1.40	1.80
Industry 4.0 and the Inter- net of Things	M.Sc. Digital Transformation Management	2023	30	1.30 (Unofficial)	1.90 (Unofficial)

References

- [1] BEKAVAC, L., MAYER, S., AND STRECKER, J. QR Code Integrity by Design. In *ACM CHI Conference on Human Factors in Computing Systems (Late-breaking Work)* (2024), ACM.
- [2] BEKTAŞ, K., STRECKER, J., MAYER, S., GARCIA, D. K., HERMANN, J., JENSS, K. E., ANTILLE, Y. S., AND SOLÈR, M. GEAR: Gaze-enabled augmented reality for human activity recognition. In *Proceedings of the 2023 Symposium on Eye Tracking Research and Applications* (2023), ACM.
- [3] GRAU, J., MAYER, S., STRECKER, J., GARCIA, K., AND BEKTAŞ, K. Gaze-based Opportunistic Privacy-preserving Human-Agent Collaboration. In *ACM CHI Conference on Human Factors in Computing Systems (Late-breaking Work)* (2024), ACM.
- [4] JENSS, K. E., AND MAYER, S. QRUco: Interactive QR Codes through Thermoresponsive Embeddings. In *Extended Abstracts of the 2023 ACM CHI Conference on Human Factors in Computing Systems* (2023), ACM.
- [5] PANDJAITAN, A., STRECKER, J., BEKTAS, K., AND MAYER, S. AuctentionAR - Auctioning Off Visual Attention in Mixed Reality. In *ACM CHI Conference on Human Factors in Computing Systems (Late-breaking Work)* (2024).
- [6] ROT, R., MAYER, S., AND STRECKER-BISCHOFF, J. Ad-hoc action adaptation through spontaneous context. In *Companion Proceedings of the the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (2025), ACM.
- [7] SCHULTZ, K., BEKTAS, K., STRECKER-BISCHOFF, J., AND MAYER, S. Open your Eyes: Blink-induced Change Blindness while Reading. In *Companion Proceedings of the the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (2025), ACM.
- [8] SEGER, D., GARCIA, K., STRECKER-BISCHOFF, J., AND MAYER, S. Dogs Go Pods: Context-dependent Access Control Rules for Sharing Personal Data of Humans and Pets. In *Companion Proceedings of the the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (2025), ACM.